

Uber: Process Mining to Optimize Customer Experience and Business Performance 10

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Abstract

Process Mining has allowed Uber's Customer Support teams to uncover insights across their processes that touch more than 700 cities across 65 countries on 6 continents. This capability allows Uber to understand variation in customer support and target large-scale multimillion dollar efficiency gains through process harmonization and increased customer satisfaction through global process benchmarking. Internally, Uber has used the power of Process Mining to help foster a culture of continuous improvement by providing a deeper level of business process visibility.

Challenge

Upon joining Uber back in 2016 as the sole lead for Process Excellence in Uber's Support Operation, it became immediately obvious to me that Uber's rapid growth had resulted in high process variation leading to lack of consistency in how we handled customer support issues. This lack of consistency created additional process waste and ultimately increased cost in agent handling time and in some situations created poor customer experiences due to a variation in the customer outcomes. If we had the ability to quickly identify all our processes and uncover unnecessary variations we would ultimately handle customer issues faster with less effort, and in some cases with automation requiring no effort at all. On top of this huge variation challenge it was very difficult to obtain reliable data allowing anyone to understand how the processes are performed across our large agent footprint.

The way to really understand how process is performed was to "go see." By observing a sample of agents performing a given process, we aimed to infer process

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improvement opportunities. Not only was this approach not scalable, it was not statistically significant. Obtaining data on a small set of agents would never provide the reliable data to support large-scale change. This lack of data that supported recommendations often led to subjective discussions about data integrity and why a given process shouldn't change. To compound this challenge we also had constraints in obtaining "accurate" data since this capability was limited to specialized analytics teams. When this constraint exists, one must be either be very influential or you must simply wait until the analytics team have the bandwidth for the ad hoc data requests. In a company with so much scale, it was imperative for the Process Excellence function to obtain data at a higher frequency in a larger volume across all cities, countries, subregions, and mega-regions.

Use Case

Community Operations support footprint spans many thousands of agents with large-scale operations in several countries covering the companies need to provide support to customers across the globe. The use case deployed for Process Mining in Uber is to identify opportunities to improve the e2e process of customer support contact handling with the goal of creating efficient (faster response) and harmonized processes that lead to fewer response errors or rework loops. All with the aim of providing a consistent high-quality service to all of our customers and partners 100% of the time.

What does the customer experience look like and where are we focusing?

In order to illustrate our use case with additional clarity, it's important to explain the customer journey and where we are applying Process Mining. Customer support within Uber deals with a vast array of customer questions across pre-trip (onboarding, setting up accounts, etc.), on-trip (Live Eats orders, trip safety, etc.) and post-trip (fare issues, navigation). In all scenarios it's imperative that we provide a fast and correct outcome. For this to happen we must have full transparency of our operational processes. Traditional reporting methods, such as scorecards, only go so deep in terms of uncovering process inefficiencies. They typically notify someone when a key metric degrades. Process Mining allows us to uncover unknown process wastes that scorecard metrics wouldn't uncover. Such issues like inefficient contact handling due to multiple hand-offs and typical LEAN transport wastes.

In terms of the implementation method used for Process Mining, we chose to implement our solution with a direct connection to our data warehouse, meaning we could push the required data on a daily basis and not worry about having to pull data into our tool when required. This approach meant that anyone in the organization could have access to the same data and insights without having Process Mining as a specialized capability within the Process Excellence team (making powerful insights more accessible enabling a continuous improvement culture).

In order to gain the maximum benefit out of Process Mining, we had to think about an internal support function that would ensure (1) the data pipelined into the tool didn't break and (2) the dashboards we designed were functional and met the